

Live Data Visualisation

CreaTe - Internet Technology – 12/06/2026

Linn Pruchnow – s35126309

Stan Fruijtier – s3515346

Noor Guit – s3460398

Lieke Aarts – s3453324



Figure 1 – Final Installation

Project Description

This project consists of a physical scale that compares two receivers that continually receive ADS-B (Automatic Dependent Surveillance-Broadcast) data. The scale ‘weighs’ the strength of the signals received by each receiver and proportionally tips towards the side representing the receiver with the highest accumulative signal strength.

ADS-B (Automatic Dependent Surveillance-Broadcast) data contains information about passing through aircrafts, such as their position, speed, altitude, and identity. The two receivers are set up on the campus of the “University of Twente” and continuously receive this information, as well as the RSSI which represents the strength of these received signals in dBm.

The data that is used to compare the two receivers can be filtered in different ways, allowing the user to gain novel insights.

By default, the scale weighs the total received signal strength [dBm] of the receivers against each other. Additionally, the user can:

- Compare the emitted signal strength of all received planes
- Average the signal strength by number of planes

The scale is a physical 3D printed model and contains a screen which displays the signal strength the receivers currently receive and the number of planes they are receiving from, as well as light-up buttons to turn the different modes on and off (figure 2).



Figure 2 - Final Installation

How It Works

The ADS-B data received by both receivers is retrieved with the provided API, used are:

- The sent address and timestamp, for keeping track of the planes individually.
- The altitude, latitude and longitude, for calculating the distance between the plane and the receiver.
- The receiver's name, for categorizing data by receiver.
- The RSSI (signal strength).

This retrieved data is sent over to a python script where the data is processed. The above data types are stored per plane. A plane will be deleted from this dataset if they are not detected within 60 seconds. After the data is processed, it is sent to the WEMOS D1 Mini microcontroller using a serial connection. The data sent per receiver is:

- The total signal strength in dBm, summed over all the current detected planes.
- The number of planes that is currently being detected.
- An estimation of the emitted signal strength per plane.

The distance between the plane and receiver is calculated using the altitude, latitude and longitude of the plane and the receiver (this information was found based on the provided location of the receiver). This distance, frequency of the emitted signal and the received RSSI are used to estimate the emitted signal strength:

$$strength_{emitted}(dBm) = RSSI + 20 * \log_{10}(1090 * 10^6 * distance) [1]$$

Where $1090 * 10^6$ is the frequency in Hz of the signal from the plane to the receiver.

This is a converted version of:

$$P(mW) = 10^{\frac{RSSI}{10}} * (f * d)^2 [2]$$

To convert formula 2 into the unit [dBm] we took $10 \log_{10}(formula\ 2)$ and the result was formula 1.

Formula 2 is based on that fact that The Friis equation shows that received power decreases approximately with $\frac{1}{(fd)^2}$.

Electronics

The heart of the scales was a Wemos D1 Mini microcontroller with a TFT-shield mounted onto it to connect to the screen. The rest of the components were wired according to the circuit diagram (figure 3). Communication between the Wemos and laptop was done through a serial connection with a USB cable. The serial connection requires GPIO pins 1 and 3 to remain unused.

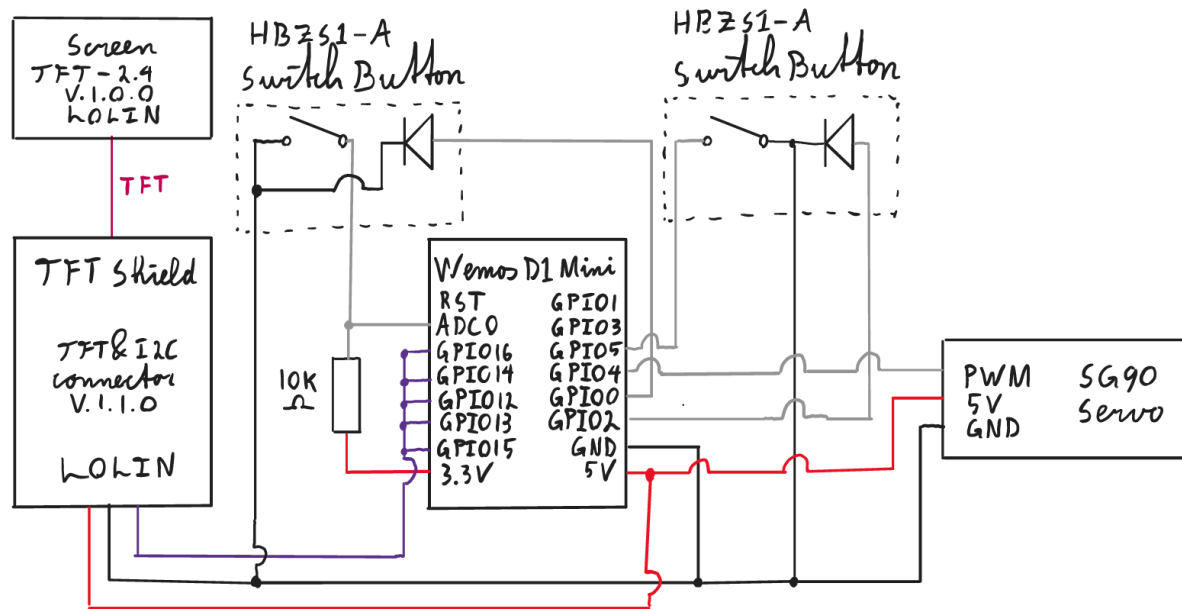


Figure 3 – Circuit Diagram

The strengths received by the dishes, or the – as previously described – estimated emitted signal strengths, are weighed against each other and mapped onto the servo position to emulate the functionality of a physical scale. The servo position was determined with the following section of code (figure 4). This code calculates the relative difference between the two strengths and translates it to a constrained servo position.

```
float a = float(Zi_5110_strength_local);
float b = float(Zi_5067_strength_local);

float score = (a - b) / (abs(a) + abs(b) + 1.0f); // outputs -1 to +1

float pos_local = 90.0f + score * 70.0f; // from neutral add excessive movement
servoWantedPos = int(constrain(pos_local, 30, 160)); // limit servo range
```

Figure 4 - Code for Servo Mapping

3D Model

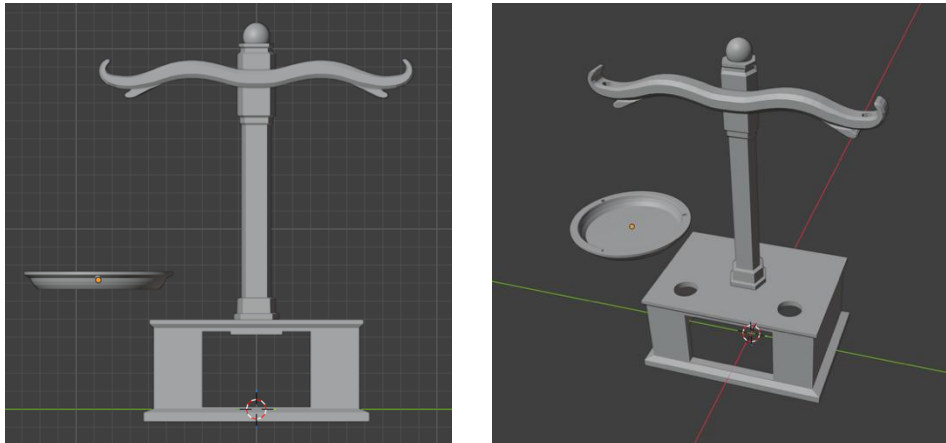


Figure 5 - Model in Blender

The model for the scale was created using Blender and adjusted to house all necessary electronics and enable the intended interactions. The final model was exported as a STL file and 3D-printed with matte PLA filament.

Why It Was Picked

We were tasked to creatively analyse network data and were intrigued by the option to work with the ADS-B data received by two differently oriented receivers. Between the LoRa and ADS-B live data, ADS-B spoke to us more as flight data was more conceptualizable with the tracking of planes being closer to the everyday experience. The plane data also provided the option to work with an entirely new type of data, the signal strength of sensor data. This data is unlike normal sensor output, with which we had already worked before.

The most obvious network data in this scenario was the signal strength of received messages, and we thought it would be interesting to compare the two receivers. Therefore, we looked for a creative way to compare received signal strength and landed on a tipping scale.

We chose this visualisation because a scale is familiar and intuitive as a method of comparing two values of the same unit. Due to the novel use case, it should still hold some intrigue.

Reflection

Learnings

We learned about ADS-B data, information about themselves when in flight, which planes constantly send out and how far reaching this signal can be. This was also the first time we worked with a live data stream we did not create ourselves but was instead shared and accessed through an API.

We had to investigate how the coordinate system works so we could translate longitude, latitude and altitude into a metric distance. We did this by first finding the distance between the ground coordinates of the plane and the Zilverling building, which is not as simple as it would be on a square grid since we live on a globe. Then we applied the Pythagorean theorem to find the distance the signal had travelled.

Because we used a microcontroller and shield without pre-resoldered headers, we had to learn how to solder these headers onto them. Due to the many pins that needed soldering we got a decent bit of practice and were quite good at it by the end.

In order to know how to estimate the emitted signal strengths, we learned about the meaning of received RSSI measurements and how to utilise them. We also learned how to use the RSSI and the distance a signal travelled to come to the estimation of the emitted signal strength.

Group Work

We divided the work in a way where everyone worked towards their strengths. Roughly the roles were design, electronics, data receiving and data processing, with the smaller roles starting on the report. This worked really well for us, as everyone could work on the parts they were familiar with in their own time.

We had a few minor issues, like a slight modelling mistake which resulted in one of the buttons not fitting properly behind the screen and sticking out. This was solved with a bit sanding of the 3D-print. There were more imperfections in the merging of the 3D printed parts and the electronics. Namely, the way the servo was mounted is fragile and the base of the model needed to be sanded for the screen to fit in its intended hole. These issues could have been prevented by paying more attention to the tolerances.

AI Statement

During the preparation of this work the authors used no artificial intelligence tools.